AI Wealth Navigator

CI/CD Pipeline Documentation

This document describes the simple automated CI/CD pipeline used for the hackathon React application.

1. Pipeline Flow

```
Developer
   ↓
GitHub Repository
   ↓
GitHub Actions
   ↓
Install Dependencies
   ↓
Build React App
   ↓
Validate dist/
   ↓
AWS S3
   ↓
Live Website
```

2. What the Pipeline Does

• Runs automatically when code is pushed to main or a pull request is opened.
• Installs the exact npm dependencies using npm ci.
• Builds the React/Vite application using npm run build.
• Checks that the production dist/index.html file exists.
• On a successful push to main, uploads the production build to AWS S3.

3. Required GitHub Secrets

Secret	Purpose
AWS_REGION	AWS deployment region
AWS_ACCESS_KEY_ID	AWS deployment access key
AWS_SECRET_ACCESS_KEY	AWS deployment secret
AWS_S3_BUCKET	Target S3 bucket name

4. Deployment Architecture

```
GitHub → GitHub Actions → AWS S3 → CloudFront → User
```

5. Security

AWS credentials are stored as GitHub Secrets and are not placed inside the source code. The deployment identity should use only the permissions required for the target S3 bucket. Real financial data and credentials must not be committed to the repository.

6. Future Extension

When the backend is implemented, additional CI/CD jobs can deploy AWS Lambda APIs, backend tests and infrastructure. The current pipeline intentionally focuses on the React frontend.

7. Submission Files

```
03_CI_CD_Pipeline/
■■■ CI_CD_Documentation.pdf
■■■ deploy.yml
```